

13. Validation / Backtesting / Paper Methodology

Evidence rules, experiment protocol, and promotion gates for policy changes

Project: QuantPipeline

Document status: Working baseline for product and engineering delivery

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Audience: Product owner, architecture, development, QA, and future implementation agents

Purpose: This document is intended to be implementable. It defines contracts, boundaries, and decisions that should survive future feature growth without breaking the core recommendation pipeline.

Document Control

Field	Value
Document code	13
Title	Validation / Backtesting / Paper Methodology
Primary input set	Documents 01-09, A1 technology recommendation, and baseline truth table
Primary architecture anchor	Data & Feature Layer -> Signals -> Selection -> Allocation -> Timing -> Risk Overlay -> Recommendation Object -> Backtest / Paper / Replay / Ops
Assumed release posture	Private single-user first release with future-safe modular boundaries
Key dependencies	Documents 10-12, 08 PRD, 09 Functional Requirements
Quality rule	No silent degradation, explicit trust semantics, and replayable output

1. Purpose of the Methodology

This document defines how QuantPipeline earns trust before and after a policy or model change is allowed to influence live recommendations. It covers historical backtesting, walk-forward validation, paper shadowing, replay linkage, promotion gates, and evidence packaging.

The methodology is designed for a medium-term investment decision product. It is not a high-frequency execution research framework. The goal is to verify that decision logic remains coherent, interpretable, and operationally honest across changing regimes, imperfect data, and alternative baselines.

2. Validation Philosophy

- Use one canonical recommendation contract across backtests, paper, replay, and current production.
- Evaluate complete policy bundles, not isolated headline metrics only.
- Freeze inputs, configuration, and code references for every serious experiment.
- Prefer walk-forward realism over single-window optimization theater.
- Treat turnover, costs, and drawdown behavior as first-class results.
- Use ablations to understand what each evidence family actually contributes.
- Require operational evidence in addition to performance evidence.
- Do not promote a policy bundle that cannot be explained or replayed.

3. Methodology Overview

Validation, Backtest, and Paper Methodology

Document 13 anchor diagram

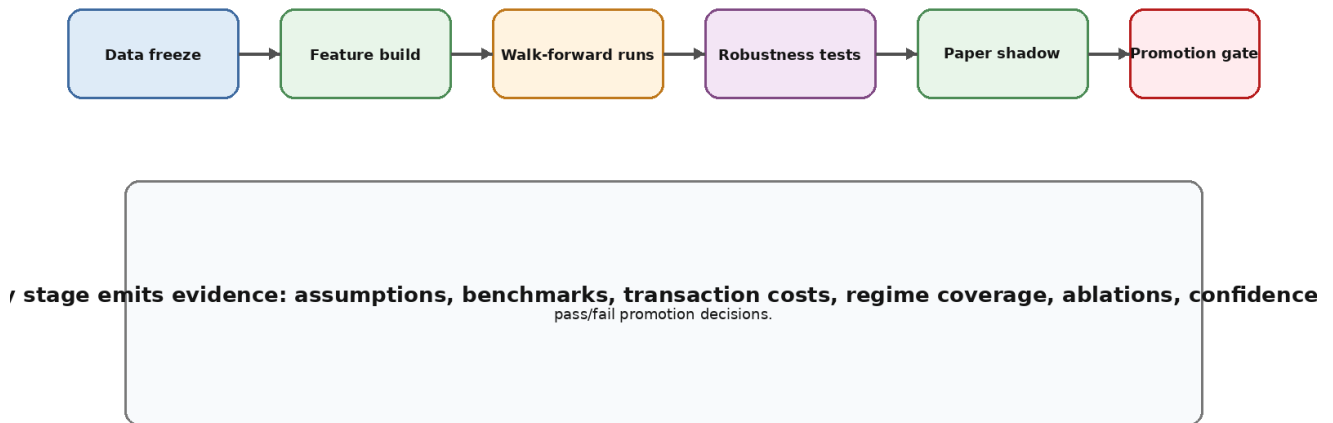


Figure 13-1. Validation and promotion pipeline.

Validation should be treated as a pipeline, not a single report. Each stage reduces a different type of risk: data freeze guards against hidden lookahead and drift, walk-forward runs test behavior over time, robustness tests challenge fragile assumptions, paper shadowing exposes operational reality, and promotion gates convert evidence into a clear yes/no decision.

4. Experiment Unit and Evidence Package

The atomic unit of evaluation is an Experiment Run. An experiment run should reference a frozen input snapshot, a policy bundle, a methodology version, cost assumptions, benchmark set, and output artifact bundle.

An Evidence Package should be generated for every promotion-grade experiment. It should include methodology parameters, period coverage, summary metrics, benchmark comparisons, drawdown analysis, turnover analysis, ablation results, regime results, paper-shadow linkage if available, and pass/fail results against promotion thresholds.

Evidence package element	Why it matters
Frozen snapshot reference	Proves what data was available and prevents quiet restatement
Policy bundle reference	Ties results to exact selection/allocation/timing/risk versions
Methodology version	Explains split rules, windows, cost assumptions, and metrics
Benchmark set	Prevents performance claims without context
Robustness outputs	Shows whether the result survives reasonable

Evidence package element	Why it matters
	perturbations
Promotion checklist	Converts analysis into a binary governance outcome

5. Data Freeze and History Segmentation

Each validation run should begin with a frozen historical dataset definition. The freeze must specify the covered universes, asset mapping rules, benchmark series, raw source versions or manifests, feature recipe versions, and event alignment logic.

History should be segmented into training, validation, and test windows using walk-forward or rolling windows rather than one static in-sample / out-of-sample split. Regime tagging should be preserved so results can be examined in expansion, contraction, stress, and recovery environments.

Window type	Role	Recommended rule
Training	Fit model components or tune thresholds	No publication claims from this window alone
Validation	Select among candidate bundles	Used for controlled comparison, not final bragging
Test	Estimate unseen performance	Strictly held out until promotion review
Walk-forward slices	Repeated realistic evaluation	Advance windows through time using frozen data at each step

6. Baselines and Benchmarks

QuantPipeline should never present a policy bundle only against its own optimized variant. Every serious evaluation needs practical baselines and recognizable benchmarks so the user can judge whether complexity is buying anything useful.

At minimum, compare against: benchmark buy-and-hold, equal-weight shortlist, simple risk-budgeted allocation, and a conservative timing or de-risking baseline. Additional baselines should be added whenever a new policy stage claims to add value.

Baseline family	Example	Purpose
Market benchmark	SPY or relevant benchmark basket	Context for directional performance
Simple portfolio policy	Equal-weight shortlist	Tests whether complex allocation truly helps
Risk-aware baseline	Vol-targeted equal-weight	Separates value from simple de-risking
No-timing baseline	Allocation without timing controller	Tests timing contribution
No-news or no-social ablation	Feature-family removal	Tests evidence-family value

7. Cost, Slippage, and Capacity Assumptions

Because the system is designed for medium-term decisions, transaction costs and turnover must be modeled consistently. Costs should include commissions or spread assumptions where relevant, and slippage should reflect the expected rebalancing cadence rather than intraday trading fantasies.

Every experiment result should store the cost model identifier used. Results that omit costs may be useful diagnostically but should never be eligible for promotion decisions.

Assumption group	Required fields
Trading friction	commission_bps, spread_bps, slippage_model_id
Rebalance policy	rebalance_frequency, threshold_rebalance_rule
Capacity	max_position_notional, liquidity_screen, participation_limit if used
Cash handling	cash_yield_assumption, idle_cash_rule

8. Robustness Test Suite

Promotion-grade experiments should run a standard robustness suite. The point is not to exhaust every imaginable variant. The point is to reject fragile bundles that only look good under one narrow framing.

Robustness test	Question answered	Promotion relevance
Regime slice analysis	Does performance vanish in specific environments?	High
Ablation analysis	Which stage or evidence family actually adds value?	High
Cost sensitivity	Does the edge survive realistic friction changes?	High
Universe sensitivity	Does the bundle overfit one narrow universe?	Medium to high
Rebalance sensitivity	Is performance dependent on one arbitrary cadence?	Medium
Data degradation simulation	Does the bundle fail honestly under missing or stale evidence?	High

9. Paper Portfolio Methodology

Paper shadowing is the bridge between historical evidence and operational trust. The paper portfolio should consume the same recommendation contract used by the overview and replay

surfaces, but it should apply explicit execution assumptions and keep its own history of rebalances, holdings, drift, and deviations.

Paper results should be comparable to backtests but not conflated with them. Backtests answer what the policy would have done on frozen history. Paper answers what happened when the live production pipeline generated recommendations under real operational conditions.

Paper object	Purpose	Must link to
paper_portfolio	Named shadow portfolio	policy bundle and benchmark
paper_rebalance	Execution of one paper action	recommendation_id and execution assumptions
paper_holding_snapshot	Observed current paper holdings	paper_portfolio_id and timestamp
paper_drift_event	Unexpected drift or discrepancy	linked paper_rebalance or external cause

10. Replay Linkage

Every promoted experiment and every paper rebalance should be linkable to replay artifacts wherever feasible. The same policy bundle that looks good statistically must also be explainable stage by stage. This reduces the risk of endorsing a method that cannot be reasoned about when it later behaves strangely.

Recommended rule: replay support is not optional for promoted policy bundles. If a bundle cannot be reconstructed in the replay model, its promotion should be blocked or explicitly exception-approved.

11. Promotion Thresholds

Thresholds should be explicit and multidimensional. A policy bundle should not pass because one return metric looks good while drawdown, turnover, or operational behavior are unacceptable.

Threshold family	Example pass rule	Fail examples
Return context	Outperform chosen baseline net of costs over test window	Only beats baseline gross of costs
Risk	Drawdown and volatility within approved envelope	Excessive drawdown concentration
Turnover	Turnover consistent with medium-term product posture	Hyperactive rebalancing
Robustness	No critical failure in standard robustness suite	Performance collapses under mild cost increase
Operational	Paper shadow and replay support available	Missing replay or repeated publication incidents

12. Reporting Pack

Every serious run should produce a consistent reporting pack so that comparisons across time remain intelligible. The report should include summary metrics, benchmark comparisons, period charts, turnover charts, drawdown analysis, contribution by decision stage if available, and a final promotion checklist.

Reports may be rendered as PDFs or dashboard pages, but the underlying raw metrics should also be stored in machine-readable form for future re-analysis.

13. Exceptions and Methodology Change Control

Methodology exceptions should be rare and documented. If a promotion is granted despite failing one threshold, the approval record must state the exception reason, compensating controls, owner, and review expiry.

Any change to windowing, benchmarks, cost model, or metric definitions should create a new methodology version. Historical results should not be silently reinterpreted under a changed methodology label.

14. Acceptance Checklist

1. Experiment references a frozen input snapshot and exact policy bundle.
2. Net-of-cost results and baseline comparisons are available.
3. Robustness suite completed and no critical failures remain unexplained.
4. Paper linkage is present where required by governance stage.
5. Replay support exists for the promoted bundle.
6. Promotion decision and rationale are recorded in admin/audit systems.

Appendix A - Core Metrics Dictionary

Suggested standard metrics include annualized return, benchmark-relative return, drawdown, volatility, turnover, hit ratio, concentration, downside capture, exposure utilization, and confidence stability. The exact formula definitions should live in a machine-readable metrics registry referenced by the methodology version.

Appendix B - Suggested Scorecard Layout

Scorecard sections: run metadata, data window summary, benchmark set, top-line metrics, risk metrics, robustness outcomes, stage contribution notes, paper observations, replay availability, pass/fail checklist, approval notes.